SUPPLEMENT TO INDUCED PATHS, RADIUS, AND EXTREMAL NEIGHBORHOOD INDEPENDENCE IN GRAPHS WITHOUT 4-CYCLES

XINGCHEN WANG AND XINCAO XU

ABSTRACT. This supplement contains the rooted bounds, the endpoint obstructions, the high-radius constructions, the candidate pairs in the ranges $4 < \ell \leq 5$ and $5 < \ell \leq 6$, the finite-search certificate for the one-triangle lemma, and a conditional reduction for GEN_3 that the article no longer uses. Numbered statements in this file carry an S prefix. Numbered references to results proved in the article point to that file. The article's Theorems on $M(n)$, radius four, and the exact triangle-free function $f(n)$ do not depend on these arguments. Theorem 5.9 uses the certificate in Section A, checked by `certificates/gen3/verify_gen3.py`.

1. Rooted bounds of orders three and four

The arguments of this section and of Sections 2 and 3 are not used for Theorems 3.6, 4.3, 5.4, 5.6, 5.7 and 5.9 or for Corollary 5.19. The rooted bound that bears on the next open band is Theorem S1.2: it is the estimate which shows that a minimum counterexample to GEN_4 is 2-connected. The distance-layer form of the same estimate is omitted. The block-class bound and the Moore observation are kept because they delimit GEN_K, and Example S3.4 is kept because it is the explicit failure of the endpoint inequalities at order nine.

For a vertex s of a connected graph F let $p_s(F)$ be the largest order of an induced path starting at s. Write $D = d_F(s)$ and $m = e(F[N(s)])$.

Definition S1.1. For an integer $q \geq 3$, the rooted bound $\mathrm{ROOT}(q)$ asserts that every connected strong C_4-free graph F and every vertex s with $p_s(F) \leq q$ satisfy

$$S(F) \leq (q-1)(|F|-1) - (q-3)D - m.$$

The hypothesis $p_s(F) \leq q$ belongs in the statement. Without it the displayed inequality is false: if $q = 3$ and $F = C_5$, then $S(F) = 10$ and $m = 0$, while $2(|F|-1) = 8$. Lemma 5.3, already proved because the cut-vertex induction of Theorem 5.4 uses it, is the case $q = 3$.

Theorem S1.2. *Let F be connected and strong C_4-free, and let $s \in V(F)$ satisfy $p_s(F) \leq 4$. Write $D = d(s)$ and $m = e(F[N(s)])$. Then*

$$S(F) \leq 3(n-1) - D - m. \tag{1}$$

In particular $\mathrm{ROOT}(4)$ *holds. Equality holds for C_5 at any root, and for a graph formed by $k \geq 1$ copies of C_5 that meet only at the root.*

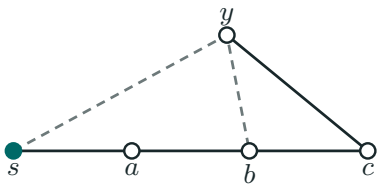
Proof. The proof is by induction on n. If $p_s(F) \leq 3$, Lemma 5.3 gives $S \leq 2(n-1) - m$. Since $D \leq n-1$, the right-hand side is at most $3(n-1) - D - m$.

Now suppose $p_s(F) = 4$, and let s, a, b, c be an induced path. The endpoint c is not a cut vertex: a neighbour of c in a component of $F - c$ that does not contain s would extend the path. Let $y \in N(c) \setminus \{b\}$. The path s, a, b, c, y is not induced, so y meets at least one of s, a, b. The

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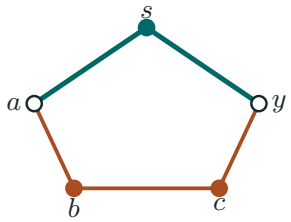
Email address of Xingchen Wang: `lovestarsni@gmail.com`.
Email address of Xincao Xu: `neard.ws@gmail.com`.

(a) A longest induced path from s



Each extra neighbour y sees s or b (dashed edges). The edge ay would create a 4-cycle.

(b) Deleting two degree-2 endpoints



If $d(b) = d(c) = 2$, delete $U = \{b, c\}$. The path a, s, y survives and $D_G(U) = 6$.

FIGURE 1. Deletion at the end of a longest induced path of order four starting at s. (a) Each neighbor $y \neq b$ of c must be adjacent to s or b. Dashed lines indicate alternatives. (b) If neither terminal vertex c nor b can be deleted with loss at most three, the indicated 5-cycle is induced and both have degree two. The deletion of $\{b, c\}$ has loss six and preserves connectivity.

edge ya would give the 4-cycle a, b, c, y. Thus y is adjacent to s or to b. The vertices s and c have at most one common neighbour, and so do b and c, hence $d(c) \leq 3$. If $d(c) = 3$, then c has two neighbours besides b, and one of them is adjacent to b. So c lies in a triangle, $t(c) \geq 1$ and $L(c) = 2d(c) - 3t(c) \leq 3$. Figure 1 shows the two deletions.

If $L(c) \leq 3$, delete c. The graph $F - c$ is connected, and $c \notin N[s]$, so D and m are unchanged. The inductive hypothesis gives $S(F - c) \leq 3(n-2) - D - m$, and therefore $S(F) \leq 3(n-2) - D - m + 3 = 3(n-1) - D - m$.

The remaining case is $d(c) = 2$ and $t(c) = 0$, so $L(c) = 4$. Let y be the neighbour of c other than b. Then y is not adjacent to b, hence ys is an edge, and s, a, b, c, y is an induced 5-cycle: every chord is excluded by the induced path s, a, b, c, by $t(c) = 0$, or by the 4-cycle that ay would create. The vertex b is the endpoint of the induced path s, y, c, b. If $L(b) \leq 3$, delete b and apply the inductive hypothesis as above.

Otherwise the same endpoint analysis gives $d(b) = 2$ and $t(b) = 0$. Delete $U = \{b, c\}$. A path that used the edges ab, bc, cy can be replaced along a, s, y, so $F - U$ is connected. The deleted edges are ab, bc and cy, and no triangle is deleted. Since $S = 2e - 3T$, one has $D_F(U) = 6 = 3|U|$. The vertices b and c lie outside $N[s]$, so D and m are unchanged. Induction gives $S(F - U) \leq 3(n-3) - D - m$, and $S(F) \leq 3(n-3) - D - m + 6 = 3(n-1) - D - m$.

For the equality examples, root C_5 at any vertex. Then $n = 5$, $D = 2$, $m = 0$ and $S = 10 = 3(n-1) - D - m$. If k copies of C_5 meet only at the root, then $n = 1 + 4k$, $D = 2k$ and $m = 0$, because the two neighbours of the root on each copy are nonadjacent, and $S = 10k = 3(n-1) - D - m$. □

Proposition S1.3. *Let F be connected and strong C_4-free, and let s satisfy $p_s(F) = 4$ and $\mathrm{ecc}(s) = 3$. Then $S(F) \leq 3(n-1) - D - m - 2$. Equality holds for P_4 rooted at an endpoint.*

Proof. The proof is by induction on n for instances with $p_s = 4$ and $\mathrm{ecc}(s) = 3$. Fix a vertex z at distance 3 from s. Whenever an endpoint other than z exists, the deletion in the proof of Theorem S1.2 applies to it. In the exceptional 5-cycle the vertices b and c are at distance 2 from s, so the deletion leaves z. The remaining graph is connected and strong C_4-free, D and m are unchanged, and distances do not decrease. Thus $d(s, z) \geq 3$. Since $p_s \leq 4$, no distance exceeds 3, so $p_s = 4$ and the eccentricity of s is still 3. A deletion of one vertex of loss at most 3 contributes 3, and the inductive bound on $n - 1$ vertices contributes $3(n-2) - D - m - 2$, which sum to $3(n-1) - D - m - 2$. Deleting $\{b, c\}$ contributes 6 against $3(n-3) - D - m - 2$.

If z is the unique endpoint, delete z. Then $p_s(F-z) \le 3$. An extra neighbour of z is not adjacent to s, for that would give $d(s,z) \le 2$. By the endpoint analysis it is adjacent to the predecessor of z and forms a triangle, so $L(z) \le 2$. Lemma 5.3 gives $S(F-z) \le 2(n-2)-m$, hence $S(F) \le 2(n-1)-m$. A geodesic from s to z has at least two vertices outside $N[s]$, so $n-1-D \ge 2$ and $2(n-1)-m \le 3(n-1)-D-m-2$.

For P_4 rooted at an endpoint one has $n=4$, $D=1$, $m=0$ and $S=6$, and the strengthened bound is $3\cdot 3-1-2=6$. □

Proposition S1.4. *Let G be a counterexample to* GEN_4 *of minimum order. Then G is* 2*-connected. This does not decide* GEN_4.

Proof. Such a graph is connected and strong C_4-free, with $p(G) \le 8$ and $S(G) \ge 4n+1$. Let v be a cut vertex, and let G_1, G_2 be the graphs of Lemma 2.3. Write ρ_i for the largest order of an induced path of G_i that starts at v. Paths from the two sides form an induced path of G on $\rho_1+\rho_2-1 \le 8$ vertices, so some side, say G_1, has $\rho_1 \le 4$. Theorem S1.2 gives $S(G_1) \le 3(n_1-1)$. The graph G_2 is a smaller connected strong C_4-free graph of path order at most 8, so minimality gives $S(G_2) \le 4n_2$. Therefore $S(G) \le 3(n_1-1)+4n_2$. Since $n = n_1+n_2-1$,

$$3(n_1-1)+4n_2 = 4n-n_1+1.$$

The hypothesis $n_1 \ge 2$ gives $4n-n_1+1 \le 4n-1$, so $S(G) < 4n$, contradicting $S(G) \ge 4n+1$. □

For $q \ge 5$ the bound $\mathrm{ROOT}(q)$ is not proved for arbitrary strong C_4-free graphs. Section 4 records the limitation. The next section proves every index for graphs whose blocks are bipartite or cycles. Conjecture S4.5 would imply $\mathrm{ROOT}(q)$ for the same q, by the attachment in Section 4. That implication is not a proof of the conjecture, and for $q \ge 5$ it does not prove $\mathrm{ROOT}(q)$.

2. Rooted bounds for every index

Let $\mathcal{B}$ be the class of connected strong C_4-free graphs in which every block is bipartite or a cycle. Edges are blocks, and cycles may be odd. The class contains every connected strong C_4-free bipartite graph and every connected strong C_4-free cactus. Theorem S2.2 keeps this block hypothesis. The graph $K_{2,3}$ does not show that the hypothesis is necessary. Its only block is itself, and that block is bipartite, but $K_{2,3}$ contains a 4-cycle subgraph. Rooted in its part of size two it has $p_s = 3$ and $S = 12 > 2(n-1)-m = 8$. The example tests the failure of $\mathrm{ROOT}(3)$ after the strong C_4-free hypothesis is removed. It does not test the block restriction inside the strong C_4-free class.

A rooted graph (G,s) with $p_s(G) = q$ and $\mathrm{ecc}(s) = q-1$ is called *terminal*. For a terminal instance put $R = n-q+1-D \ge 0$, the number of vertices outside $N[s]$ and outside a root-to-farthest-vertex geodesic. Set $c_2 = 0$, $c_3 = 2$, $c_4 = 3$ and $c_q = 2\lfloor (q-1)/2 \rfloor$ for $q \ge 5$.

Lemma S2.1. *Let G be connected and strong C_4-free, and let P be a longest induced path starting at a vertex s, of order q. Let x be the other end of P, and suppose that the block B containing x is bipartite. Let Q be the subpath of P formed by the vertices of P that lie in B, and set $\ell = |Q|$. Then every neighbour of x lies in B, and x lies in no triangle. If $\ell = 2$, then $d(x) = 1$. If $\ell \ge 3$, then $d(x) \le \lfloor (\ell-1)/2 \rfloor$. In every case $\ell \le q$, and $L(x) \le c_q$ whenever $q \ge 5$.*

Proof. A neighbour of x in a component of $G-x$ that does not contain s would extend P, so x is not a cut vertex. Every edge with one end in B and one end outside B is incident with a cut vertex of B. Therefore every neighbour of x lies in B. The block is bipartite, so x lies in no triangle.

Since the block-cut tree is a tree, P enters B only once. The intersection of P with B is a single subpath $Q = z_0, \dots, z_{\ell-1}$ with $z_{\ell-1} = x$. If s lies in B, then $Q = P$. If s lies outside B, the vertices of P that precede z_0 lie outside B, and z_0 is their only neighbour in B: any other edge

from that prefix into B would be a second entry. A vertex of $B \setminus \{z_0\}$ is therefore adjacent to no vertex of the prefix. An extension of P at x through a vertex of B can fail only by a chord into Q.

Let y be a neighbour of x other than $z_{\ell-2}$. Then y lies in B, and y does not lie on Q, because Q is induced. The vertex y is not adjacent to the prefix. If y had no neighbour on Q, the path P followed by y would be induced. Thus y has a neighbour z_i on Q. This neighbour is not x. It is not $z_{\ell-2}$, for $yxz_{\ell-2}$ would be a triangle, and it is not $z_{\ell-3}$, for $z_{\ell-3}z_{\ell-2}xy$ would be a 4-cycle. Since B is bipartite, z_i has the colour of x, so $i \equiv \ell - 1 \pmod 2$ and $i \le \ell - 5$.

If $\ell = 2$, then Q is a single edge and there is no admissible index, so $d(x) = 1$. If $\ell \ge 3$, the number of admissible indices is $\lfloor(\ell-1)/2\rfloor - 1$. Two neighbours of x are nonadjacent and cannot share a blocker, for a common blocker would form a 4-cycle with them and with x. Hence $d(x) \le \lfloor(\ell-1)/2\rfloor$. In every case $\ell \le q$. For $q \ge 5$ one has $d(x) \le \lfloor(q-1)/2\rfloor$, and $L(x) = 2d(x) \le c_q$.

The same count applies inside B. Let z be the cut vertex of B lying on every path from a fixed root to B, or the root itself if the root lies in B, and let Q be a longest induced path of B starting at z, with other end x and with $\ell = |Q|$. A neighbour of x in B that had no neighbour on Q would extend Q inside B. If the root lies outside B, no vertex of $B \setminus \{z\}$ meets the component of the root at z, so that component contributes no further blocker. Thus $d_B(x) \le 1$ if $\ell = 2$, and $d_B(x) \le \lfloor(\ell-1)/2\rfloor$ if $\ell \ge 3$. □

Theorem S2.2. (1) *Every terminal instance in $\mathcal{B}$ satisfies $S(G) \le 2(D+q-2) - m + c_q R$.*

(2) *Every $F \in \mathcal{B}$ with a vertex s such that $p_s(F) \le q$ satisfies*

$$S(F) \le c_q(n-1) - (c_q - 2)D - m.$$

In particular ROOT(q) *holds in $\mathcal{B}$ for every $q \ge 3$. If every block is bipartite or a triangle, one may take $c_q = 2\lfloor(q-1)/2\rfloor$ for all $q \ge 3$.*

Proof. Two terminal facts. An endpoint x of a longest rooted induced path is not a cut vertex: a neighbour of x in another component of $G-x$ would extend the path. If z is at distance $q-1$ from s and $v_0 = s, \dots, v_{q-1} = z$ is a geodesic, any neighbour $y \ne v_{q-2}$ of z must meet an earlier path vertex (else the rooted path extends); meeting v_i with $i \le q-4$ shortens $d(s,z)$, and meeting v_{q-3} gives a C_4; so $y \sim v_{q-2}$, and two such y would give a C_4. Hence $L(z) = 2d(z) - 3t(z) \le 2$. Moreover, if $x \ne s, z$, $G-x$ is connected and z survives, then $q-1 \le d_{G-x}(s,z) \le p_s(G-x) - 1 \le q-1$, so the instance stays terminal with the same q.

Endpoint losses. Let x be the endpoint of a longest rooted path of order q. It lies in a single block. If that block is a triangle the loss $L(x)$ is 1, if an edge 2, and if a cycle of length at least 5 it is 4. If the block is bipartite, Lemma S2.1 gives $L(x) \le c_q$ for $q \ge 5$.

No nontrivial bipartite block when $q \le 4$. Let B be a nontrivial bipartite block, and let z be the cut vertex of B that lies on every path from s to B, or let $z = s$ if $s \in B$. Let Q be a longest induced path of B starting at z, and let x be its other end. Every vertex of a nontrivial block has degree at least two in the block, so $d_B(x) \ge 2$. The last paragraph of Lemma S2.1 gives $d_B(x) \le 1$ if $|Q| \le 4$ and $d_B(x) \le \lfloor(|Q|-1)/2\rfloor$ in any case with $|Q| \ge 3$. The inequality $d_B(x) \ge 2$ therefore forces $|Q| \ge 5$. An induced s–z path meets B only at z. Concatenating it with Q yields an induced path that starts at s. The order of this path is $|Q|$ if $s \in B$, and at least $|Q|+1$ if $s \notin B$. In either case $p_s(G) \ge 5$. A terminal instance with $q \le 4$ therefore has no nontrivial bipartite block. Every block is an edge, a triangle, or a cycle of length at least five, so the graph is a cactus.

Bases. For $q \le 4$ the preceding paragraph leaves only cacti. For a C_4-free cactus with c_3 triangle blocks and c_L longer cycle blocks, $S = 2(n-1) + 2c_L - c_3$ and $m \le c_3$. For $q = 3$ no longer cycle occurs, and (1) follows. For $q = 4$ every longer cycle is a C_5 through s. A chosen length-3 geodesic uses at most one edge of each such C_5: if it used two edges and then left the cycle, replacing that

two-edge arc by the other three-edge arc would give an induced rooted path of order 5. So each C_5 contributes two vertices to R, $R \geq 2c_L$, and $[2(D+2) - m + 3R] - S = R - 2c_L + c_3 - m \geq 0$. For $q = 2$ the root is universal and $S = 2D - m$.

Induction on n. Let E be the set of endpoints of rooted induced paths of order q and z a farthest vertex, so $z \in E$. If some $x \in E \setminus \{z\}$ exists, delete it: $x \notin N[s]$, x is not a cut vertex, its loss is at most c_q, the instance remains terminal with the same q, D and m, and R drops by one. If $E = \{z\}$, delete z: the new rooted maximum is $q-1$, the eccentricity $q-2$, R is unchanged and the loss is at most 2; induction gives $S \leq 2(D+q-3) - m + c_{q-1}R + 2 \leq 2(D+q-2) - m + c_qR$. Deletion preserves membership in $\mathcal{B}$. This proves (1); the bipartite-or-triangle variant is identical.

(2) Attach a new path with $q-1$ edges at s. The result is terminal of index q, lies in $\mathcal{B}$, has order $n+q-1$, mass $S + 2(q-1)$ and root degree $D+1$, and m is unchanged. Substituting into (1) gives (2). Since $c_q \leq q-1$ and $n - 1 - D \geq 0$,

$$\begin{aligned} c_q(n-1) - (c_q - 2)D - m &= c_q(n - 1 - D) + 2D - m \\ &\leq (q-1)(n-1-D) + 2D - m \\ &= (q-1)(n-1) - (q-3)D - m, \end{aligned}$$

so ROOT(q) follows. □

A matching upper bound of shape $S \leq (r-1)n - 2r(r-3)$, for graphs in $\mathcal{B}$ of radius $r \geq 5$ and path order $2r-1$, has been proposed from a structural description of radius-extremal graphs. That description is not proved in this paper, and the radius bound in $\mathcal{B}$ is not claimed.

Lemma S2.3. *Let G be strong C_4-free and triangle-free, and let $x_0x_1 \cdots x_k$ be an induced path that cannot be extended at x_k. If $d(x_k) \geq 2$, then $k \geq d(x_k) + 1$. If moreover G is bipartite, then $k \geq 2d(x_k)$.*

Proof. Let $y \neq x_{k-1}$ be a neighbour of x_k. It is not on the path, and, since the path cannot be extended, y has a neighbour x_i with $i \leq k-1$. Here $i \neq k-1$ (a triangle) and $i \neq k-2$ (the 4-cycle $x_{k-2}x_{k-1}x_ky$), so $i \leq k-3$. A vertex x_i is adjacent to at most one neighbour of x_k. Hence the $d(x_k) - 1 \geq 1$ such neighbours need distinct blockers among $x_0, \ldots, x_{k-3}$, and $k - 2 \geq d(x_k) - 1$. If G is bipartite, the blocker has the colour of x_k, so $i \equiv k \pmod 2$ and $i \leq k-4$; there are $\lfloor k/2 \rfloor - 1$ such positions, and $\lfloor k/2 \rfloor \geq d(x_k)$. □

Proposition S2.4. *Every connected regular bipartite strong C_4-free graph satisfies* GEN_K *for every K and satisfies REAL-133. Every Moore graph of diameter* 2 *and girth* 5 *satisfies REAL-133.*

Proof. Let G be d-regular. If $d = 1$ then $G = K_2$, with $r = 1$, $\ell = 1$, $p = 2$ and $S = n$, and both statements hold. Let $d \geq 2$. As G is triangle-free, $\lambda \equiv d$ and $\ell = d$. Lemma S2.3 applied to a longest induced path gives $p \geq 2d + 1$, and Theorem 2.4 gives $p \geq 2r - 1$; so $p \geq r + d$ both when $r \leq d + 1$ and when $r \geq d + 1$. If $p \leq 2K$, then $2d + 1 \leq 2K$, so $S = dn < Kn$. A Moore graph of diameter 2 and girth 5 has degree $d \geq 2$, radius 2 and $\ell = d$. The degree is an integer, so the floor form and the real form ask for the same inequality. The first part of Lemma S2.3 gives $p \geq d + 2 = r + \ell$. This is an alternate proof of the case already recorded by Chen [3] immediately after his Theorem 3.6. It is not a new extremal classification. □

Remark S2.5. A Moore graph of degree 57, if it exists, has $n = 57^2 + 1 = 3250$ and $S = 57n$. Proposition S2.4 gives REAL-133, which only needs $p \geq 59$, and this conclusion is the integer case of the fact cited above. The statement GEN_{57} holds for it with equality in the mass bound, but GEN_{56} asks for $p \geq 113$ because $S > 56n$. We do not know whether such a graph would have $p \leq 112$. If it did, it would refute GEN_{56} while satisfying REAL-133. Its existence would therefore not by itself separate the two statements, but it shows that a proof of every GEN_K is a different assertion from REAL-133.

3. Negative results

The examples in this section are explicit graphs.

Example S3.1. Let $s \geq 3$, $t \geq 3$ and $m \geq 1$ be integers. Let $G_{s,t,m}$ be the cycle $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of t edges at every cycle vertex, together with m internally disjoint paths c_0xzyc_1. Then $n = 2s(t+1) + 3m$, $e = 2s(t+1) + 4m$ and $T = 0$, so $e - n - T = m \geq 1$. For $(s,t,m) = (3,3,1)$ one has $n = 27$, $e = 28$, $r = 6$ and $p = 11$. These graphs satisfy REAL-133, since $\ell = 2e/n < 8/3$. They also show that $e = n + T$ fails for radius-extremal strong C_4-free graphs of every radius $r \geq 6$: here $r(G_{s,t,m}) = s + t$ and $p(G_{s,t,m}) = 2(s+t) - 1$.

Two vertices have at most one common neighbour, so G is strong C_4-free and triangle-free. An added path of length four does not shorten distances among the older vertices, and c_0 remains within distance two of every vertex of such a path, so the eccentricity of c_0 is still $s+t$. A vertex w of an added path meets the tip of the arm at c_s only through c_0 or c_1, hence at distance at least $s+t$. Thus $r = s+t$. For the path order, let B be the block formed by C and the added paths. An induced path P meets B in a subpath Q and can leave B only at an end of Q that is a cycle vertex, through at most t arm vertices. The same case division as for a single ear shows $|P| \leq 2s + 2t - 1$, using $t \geq 3$ and $s \geq 3$, and the cycle minus one vertex together with the arms at its two ends attains this order.

The last sentence of Example S3.1 abbreviates a case division which we now record. If Q avoids both c_0 and c_1, then $|Q| \leq 2s-2$. If Q contains exactly one of them, then either Q lies on the cycle with $|Q| \leq 2s-1$, or $|Q| \leq 2s+2$ with at most one cycle end, or $|Q| \leq 7$ with no cycle end. If Q contains both, the same three shapes occur with $|Q| \leq 2s-1$, $|Q| \leq 2s+2$ and $|Q| \leq 6$. Adding at most t arm vertices at each cycle end of Q gives $|P| \leq \max\{2s-1+2t,\ 2s+2+t,\ 7\} = 2s+2t-1$.

Example S3.2. Forbidding induced paths on eleven vertices and requiring $L(v) \geq 6$ at every vertex does not forbid an induced star $K_{1,6}$. Two Petersen graphs sharing a vertex give a triangle-free strong C_4-free graph on $n = 19$ vertices with $S = 60$, minimum deletion loss 6, $p = 9$ and $\lambda = 6$ at the shared vertex. For $b \geq 2$, gluing b copies of the Petersen graph along a common edge gives a 2-connected graph B_b with $n = 8b+2$, $S = 28b+2$, minimum loss 6, $p = 9$ and $\lambda = 2b+1$ at the two hubs. Thus singleton losses do not bound λ. These graphs are not counterexamples to GEN_5: the eight non-hub vertices of one Petersen block have deletion loss $28 < 5 \cdot 8 + 1$.

For an oriented edge (u,v) of a connected strong C_4-free graph G, let q be the maximum order of an induced path whose first two vertices are u, v in that order, and let E be the set of terminal vertices of the induced paths of that order. The endpoint inequality $\mathrm{EP}(q)$ asserts $D_G(E) \leq q|E|$ for every such oriented edge of maximum order q. For a root s with $p_s(G) = q$, if E is the set of terminal vertices of the induced paths of order q starting at s, the rooted inequality $\mathrm{EP}^+(q)$ asserts $D_G(E) \leq (q-1)|E|$.

Proposition S3.3. *For every integer $q \geq 4$, $\mathrm{EP}(q)$ holds for all connected strong C_4-free graphs if and only if $\mathrm{EP}^+(q-1)$ does.*

Proof. Both graphs below remain strong C_4-free, so $S = 2e - 3T$ and the deletion loss of a vertex set is determined by the edges and triangles that meet it.

First assume $\mathrm{EP}^+(q-1)$, and let (u,v) be an oriented edge of maximum induced-path order q, with terminal set E. Delete $W = N[u] \setminus \{v\}$, and let H be the component of v in $G - W$. An induced path of G that starts with uv has no further vertex in $N[u]$, because the path is induced, so the remainder is an induced path of H starting at v. Conversely, every induced path of H starting at v extends by u to an induced path of G, since no vertex of H other than v is adjacent to u. Thus $p_v(H) = q-1$, and the terminal set in H is still E.

Each $x \in E$ has at most one neighbour in W: any such neighbour is adjacent to u, and two of them would be two common neighbours of u and x. Every neighbour of x outside H lies in W.

Let m_W be the number of edges between E and W, so $m_W \le |E|$, and let T_{extra} be the number of triangles that meet E and are not contained in H. Every such triangle uses one of those edges. Comparing edges and triangles,

$$D_G(E) = D_H(E) + 2m_W - 3T_{\text{extra}} \le D_H(E) + 2|E|.$$

The rooted inequality in H is $D_H(E) \le (q-2)|E|$, because $\mathrm{EP}^+(q-1)$ contributes the factor $(q-1)-1$. Therefore $D_G(E) \le q|E|$.

Conversely, assume $\mathrm{EP}(q)$, and let (H, v) be a connected strong C_4-free graph with $p_v(H) = q-1$ and terminal set E. Form G by adding a vertex u adjacent to v and, for each $x \in E$, a new vertex a_x adjacent exactly to u and to x. Every terminal x fails to be adjacent to v: an induced path of order $q-1 \ge 3$ from v to x has no chord vx. The vertex a_x has degree two. Any cycle through a_x uses both edges ua_x and a_xx, and then continues from x to the other neighbour of u on the cycle. That other neighbour is v or some a_y. In the first subcase the x–v path in H has length at least two, so the cycle has length at least five. In the second the cycle has the form $u, a_x, x, \ldots, y, a_y$ and again has length at least five. A cycle through u that avoids every a_x would need two neighbours of u in H, but the only neighbour of u in H is v. Thus G is strong C_4-free. No new triangle appears: the two neighbours of a_x are nonadjacent, and the neighbours of u are pairwise nonadjacent.

An induced path that starts with uv contains no a_x. Otherwise the edge ua_x would be a chord, since a_x is not the successor of u on that path. The induced uv-paths are therefore the paths u followed by an induced path of H starting at v. Their maximum order is q, and their terminal set is E. Exactly $|E|$ new edges meet E, namely the edges a_xx, and no new triangle meets E. Hence $D_G(E) = D_H(E) + 2|E|$. The oriented inequality gives $D_G(E) \le q|E|$, so $D_H(E) \le (q-2)|E|$, which is $\mathrm{EP}^+(q-1)$. □

Example S3.4. EP(9) fails. Let G have vertices $0, \ldots, 14$ and adjacency lists

0: 4, 8, 11	1: 5, 8, 10	2: 6, 8, 9	3: 7, 10
4: 0, 9, 10	5: 1, 9, 11	6: 2, 10, 11	7: 3, 11
8: 0, 1, 2, 12	9: 2, 4, 5	10: 1, 3, 4, 6, 13	11: 0, 5, 6, 7, 14
12: 8, 13, 14	13: 10, 12	14: 11, 12.	

The graph is triangle-free and strong C_4-free, with $n = 15$, $e = 23$ and $S = 46$. For the oriented edge $(12, 8)$ the maximum order is $q = 9$, attained only by

$$12, 8, 2, 9, 4, 10, 3, 7, 11 \qquad \text{and} \qquad 12, 8, 2, 9, 5, 11, 7, 3, 10.$$

Thus $E = \{10, 11\}$ and $D_G(E) = 20 > 18$. No induced path that starts with $12, 8$ has order greater than 9. The neighbours of 8 other than 12 are $0, 1, 2$. A continuation is legal only when the new vertex meets the current end and meets no earlier vertex of the path. Under that rule every maximal induced path is one of the following, and the only two of order 9 are the paths already displayed.

maximal induced path starting $12, 8$	order
$12, 8, 0, 4, 9, 5$	6
$12, 8, 0, 4, 10, 3, 7$	7
$12, 8, 0, 4, 10, 6$	6
$12, 8, 0, 11, 5, 9$	6
$12, 8, 0, 11, 6, 10, 3$	7
$12, 8, 0, 11, 7, 3, 10$	7
$12, 8, 1, 5, 9, 4$	6
$12, 8, 1, 5, 11, 6$	6
$12, 8, 1, 5, 11, 7, 3$	7
$12, 8, 1, 10, 3, 7, 11$	7
$12, 8, 1, 10, 4, 9$	6
$12, 8, 1, 10, 6, 11, 7$	7
$12, 8, 2, 6, 10, 3, 7$	7
$12, 8, 2, 6, 10, 4$	6
$12, 8, 2, 6, 11, 5$	6
$12, 8, 2, 6, 11, 7, 3$	7
$12, 8, 2, 9, 4, 10, 3, 7, 11$	9
$12, 8, 2, 9, 5, 11, 7, 3, 10$	9

The first of the two paths of order 9 ends at 11. The neighbours of 11 are $0, 5, 6, 7, 14$: the vertex 7 is the predecessor, 0 meets 8, 5 meets 9, 6 meets both 2 and 10, and 14 meets 12. The second path ends at 10. The neighbours of 10 are $1, 3, 4, 6, 13$: the vertex 3 is the predecessor, 1 meets 8, 4 meets 9, 6 meets both 2 and 11, and 13 meets 12. Neither path extends. The same computation gives a failure of $\mathrm{EP}^+(8)$ on the subgraph induced by $0, \dots, 11$, rooted at 8: $p_8 = 8$, $E = \{10, 11\}$ and $D(E) = 16 > 14$. The inequalities EP(8) and $\mathrm{EP}^+(6)$ are not settled here.

Example S3.5. Let $k = p - r$. Call an induced subgraph H of G an *eligible core* if it is proper, connected and dominating, and $p(H) \le p(G) - 2$. The budget CB asks for an eligible core with

$$S(G) - S(H) \le k\big(n - |H|\big) + |H|.$$

Together with REAL-133 for H and an induction on the order, the budget would imply REAL-133. The Petersen graph has $S = 30$, $p = 5$, $r = 2$ and $k = 3$, hence $\ell = 3$. An eligible core has $p(H) \le 3$. A connected triangle-free strong C_4-free P_4-free graph is a star, by Lemma 2.7 and the absence of triangles. In the Petersen graph a star with $k \le 3$ leaves has centre of degree 3. The vertices it dominates are the centre, its three neighbours, and at most two further neighbours of each leaf, at most $4 + 2k$ vertices in all. This is 10 only for $k = 3$, so the eligible cores are exactly the ten closed neighbourhoods $N[v]$. Each has $S(H) = 6$, and $S(G) - S(H) = 24 > 3 \cdot 6 + 4 = 22$.

The unrestricted budget therefore fails. The example has $\ell = 3$. It does not determine a cutoff such as $\ell > 5$, and this paper contains no census of eligible cores.

Example S3.6. The degree, triangle and distance data of the Heawood graph ($n = 14$, cubic, $T = 0$, $r = 3$, $S = 42$, and $Z = 28$ nonadjacent pairs with no common neighbour) satisfy the counting identities used above, including $\sum_v d(v)^2 = n(n-1) + 6T - 2Z$, together with the lower bounds $p \ge 2r - 1$ and $p \ge \min_v \lambda(v) + 2$, if p is replaced by the value 5. But $5 < r + \ell = 6$. This system of scalar constraints does not imply REAL-133. The Heawood graph itself has $p = 7$.

Example S3.7. Let F be the incidence graph of the projective plane $\mathrm{PG}(2, q)$, with $f = |F| = 2(q^2 + q + 1)$ and $S(F) = (q + 1)f$, and choose a root of F. Let $t = f - 1$ and $b \ge 1$. Take a 6-cycle $c_0 \cdots c_5$, attach a pendant path with t edges at each cycle vertex, and attach b copies of F at c_0 by identifying their roots with c_0. The result is strong C_4-free, because each new block meets the rest in a cut vertex, and

$$n = 6f + b(f - 1), \qquad S = 12f + b(q + 1)f, \qquad r = t + 3, \qquad p = 2t + 5 = 2r - 1.$$

Distances inside the flower are unchanged. Every vertex of an attached copy is within distance $f-1 \le t$ of c_0, so each cycle vertex has eccentricity $t+3$, while a vertex x of a copy has distance $d_F(x,c_0)+t+3$ from the tip of the arm at c_3. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle contributes at most t vertices. Mass is additive at cut vertices by Lemma 2.3. For fixed q, as $b \to \infty$ one has $\ell \to (q+1)f/(f-1)$. Letting $q \to \infty$, with $r = f+2$, this limit grows like $\sqrt{r/2}$. For $q = 5$ and $b = 17$ one has $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. These graphs satisfy RAD(r) for their own radius, and they show that ℓ is unbounded on the class $p = 2r-1$. Any theorem RAD(r) must use a bound that depends on r.

4. Candidate pairs above density four

The two lemmas below are the density restrictions used for the pairs $(r,p) = (2,6)$, $(3,6)$ and $(2,7)$. They are not used in the proofs of Theorems 3.6, 4.3, 5.4, 5.6 and 5.7.

Lemma S4.1. *Let G be connected, bipartite and strong C_4-free, with $r(G) = 2$. Then G is a tree, $\ell(G) = 2 - 2/n$ and $p(G) \le 5$.*

Proof. Let s be a centre, and let A be the colour class of s and B the other class. Every vertex of B lies at odd distance from s, hence at distance 1, so $B \subseteq N(s)$. Every vertex $x \in A \setminus \{s\}$ lies at distance 2 from s, so x has a neighbour in B. Two such neighbours would form a 4-cycle with s. Thus x has exactly one neighbour. There is no edge inside A or inside B, and the number of edges is $|B| + |A| - 1 = n - 1$. A connected graph with $n-1$ edges is a tree. There are no triangles, so $S = 2e = 2n-2$ and $\ell = 2 - 2/n$. Every vertex of $A \setminus \{s\}$ is a leaf. An induced path contains at most two leaves, and the unique path between them has the form leaf, neighbour in B, s, neighbour in B, leaf. Hence $p(G) \le 5$. □

Lemma S4.2. *Let G be a connected triangle-free strong C_4-free P_8-free graph. Then $e(G) \le 2n-2$. In particular, for $n \ge 2$ one has $\ell(G) = 2e(G)/n < 4$.*

Proof. The graph has no triangle and no 4-cycle subgraph, so every cycle has length at least 5. The bound $e \le 2n-2$ holds for $n = 1$. Let $n \ge 2$, and proceed by induction on the order. If G is a tree, then $e = n-1 \le 2n-2$. Otherwise the girth is at least 5. Theorem 1 of Chudnovsky and Stacho [4] then asserts that G is the Petersen graph, the Heawood graph, the graph obtained by contracting one edge of the Heawood graph, or else G has a vertex of degree at most 2. These three exceptions have pairs (n,e) equal to $(10,15)$, $(14,21)$ and $(13,20)$. Each satisfies $e \le 2n-2$, since $15 \le 18$, $21 \le 26$ and $20 \le 24$.

In every remaining case some vertex v has degree at most 2. Every component H of $G - v$ is connected, triangle-free, strong C_4-free and P_8-free. The inductive hypothesis gives $e(H) \le 2|H| - 2$. If $G - v$ has $c \ge 1$ components, then

$$e(G) = e(G-v) + d(v) \le 2(n-1) - 2c + 2 \le 2n - 2,$$

because $d(v) \le 2$ and $c \ge 1$. Since G is triangle-free, $S = 2e \le 4n-4$, so $\ell \le 4 - 4/n < 4$. □

Lemma 5.14 gives $\ell < 4$ whenever $p \le 6$. The pairs $(r,p) = (2,6)$ and $(3,6)$ are therefore excluded from the range $\ell \ge 4$, and from no wider range. Proposition 6.2 realises $(3,6)$: the graph θ_k has $\ell = 3 - 6/n$. Lemma S4.2 applies to every connected triangle-free strong C_4-free graph of induced-path order 7, since such a graph is P_8-free, and yields $\ell < 4$. Equivalently, if $p = 7$ and $\ell \ge 4$, then G contains a triangle. In the range $5 < \ell \le 6$ the pairs $(2,7)$ and $(3,7)$ therefore occur only for graphs that contain a triangle. The same conclusion does not forbid $(r,p) = (2,7)$ in a triangle-free graph. Lemma S4.1 is the bipartite restriction: a connected bipartite strong C_4-free graph of radius 2 is a tree, so its induced-path order is at most 5.

Example S4.3. Let P be the Petersen graph, and let u and v be the ends of an induced P_5 in P. Such a path exists by the proof of Theorem 5.6, and u and v are nonadjacent. Form G by adding one leaf at u and one leaf at v. Then

$$(r(G), p(G), |V(G)|, S(G), \ell(G)) = (2, 7, 12, 34, 17/6).$$

The graph P has 15 edges. The two new edges create no cycle, so G is triangle-free and strong C_4-free, $n = 12$, $e = 17$ and $S = 2e = 34$. Thus $\ell = 17/6$. The graph P is P_6-free, because it is the extremal graph of order 10 in Theorem 3.6. A leaf has degree one, so it is an end of every induced path that contains it. Deleting the leaves that lie on an induced path of G leaves an induced path of P, of order at most 5. The path formed by the two leaves and the chosen P_5 is induced, because each leaf is adjacent only to its endpoint. Hence $p(G) = 7$. The radius is 2. The proof of Theorem 3.6 records that the diameter of P is 2, so u and v have a common neighbour w. Every vertex of P is at distance at most 2 from w, and each new leaf is at distance 2 from w. Thus the eccentricity of w is 2.

Suppose $4 < \ell \le 5$ and G violates REAL-133. Then $\lceil \ell \rceil = 5$, so Corollary 6.1 with $K = 4$ gives either $r \le 4$ and $2r - 1 \le p \le r + 4$, or $r = 5$ and $p = 9$. Lemma 5.14 forces $\ell < 4$ whenever $p \le 6$, so $p \ge 7$. Corollary 4.4 forces $\ell \le 2$ when $(r, p) = (4, 7)$. The pairs that remain are $(3, 7)$, $(4, 8)$ and $(5, 9)$.

For $5 < \ell \le 6$, Corollary 6.1 gives $2r - 1 \le p \le r + 5$ and $r \le 6$. Lemma 3.1 excludes radius 1, Theorem 3.6 excludes $p \le 5$, and Corollary 4.4 excludes $(4, 7)$. The pairs $(2, 6)$ and $(3, 6)$ do not occur in this range, by the density restriction above, though $(3, 6)$ occurs at $\ell = 3 - 6/n$. If $p = 7$ and $\ell \ge 4$, then G contains a triangle, so $(2, 7)$ and $(3, 7)$ occur in this range only for graphs that contain a triangle. The pairs that remain are necessary conditions for a counterexample. The table does not assert that any of them occurs.

For $q \ge 5$ the bound ROOT(q) is not proved for an arbitrary strong C_4-free graph. Theorem S2.2 gives every index in the block class $\mathcal{B}$. Proposition S1.4 shows that a minimum counterexample to GEN$_4$ is 2-connected, and GEN$_4$ itself is open. The statements GEN$_K$ for $K \ge 4$, unrestricted GEN$_6$, RAD(r) for $r \ge 5$, and ROOT(q) for $q \ge 5$ outside $\mathcal{B}$ are open. No structure theorem for connected strong C_4-free graphs of radius $r \ge 5$ and path order $2r - 1$ is proved in the article.

Conjecture S4.4. ROOT(q) *holds for every $q \ge 3$. For every $r \ge 5$, every connected strong C_4-free graph of radius r and path order $2r - 1$ satisfies $S \le (r - 1)n - 2r(r - 3)$, with equality only for C_{2r}.*

The cases $q = 3$ and $q = 4$ are Lemma 5.3 and Theorem S1.2, and every q holds in $\mathcal{B}$ by Theorem S2.2. The radius bound is not claimed for $r \ge 5$.

Conjecture S4.5. *If F is connected and strong C_4-free, $p_s(F) = q$ and the eccentricity of s is $q - 1$, then*

$$S(F) \le (q - 1)(n - 1) - (q - 3)D - m - (q - 2)(q - 3).$$

For $q = 4$ the subtracted term is 2, and Proposition S1.3 is this case. For $q \ge 5$ the conjecture is open. The terminal bound in $\mathcal{B}$ is the first part of Theorem S2.2.

Conjecture S4.5, for a fixed q, implies ROOT(q). The implication does not prove the conjecture. Let F be connected and strong C_4-free with $p_s(F) \le q$. Form G by attaching a path of $q - 1$ new edges at s. Then G is terminal of index q: the new path is an induced path of order q starting at s, and every induced path of F that starts at s has order at most q. The order, the root degree and the mass increase by $q - 1$, 1 and $2(q - 1)$ respectively, and m is unchanged. Write $N = n + q - 1$ and $D_G = D + 1$. The terminal inequality in G reads

$$S(F) + 2(q - 1) \le (q - 1)(N - 1) - (q - 3)D_G - m - (q - 2)(q - 3).$$

Statement	Status
ROOT(3) and ROOT(4); the terminal strengthening of Proposition S1.3	proved, Lemma 5.3 and Theorem S1.2
ROOT(q) for every q, in the class $\mathcal{B}$	proved, Theorem S2.2
a minimum counterexample to GEN_4 is 2-connected	proved, Proposition S1.4; GEN_4 is open
GEN_K for $K \geq 4$, unrestricted GEN_6, RAD(r) for $r \geq 5$, and ROOT(q) for $q \geq 5$ outside $\mathcal{B}$	open
pairs (r, p) still necessary for $4 < \ell \leq 5$	$(3, 7)$, $(4, 8)$, $(5, 9)$
pairs (r, p) still necessary for $5 < \ell \leq 6$	$(2, 7)$, $(3, 7)$, $(3, 8)$, $(4, 8)$, $(4, 9)$, $(5, 9)$, $(5, 10)$, $(6, 11)$
$(2, 6)$ and $(3, 6)$ in the range $\ell \geq 4$	excluded, by Lemma 5.14; θ_k realises $(3, 6)$ at $\ell = 3 - 6/n$
$(2, 7)$ and $(3, 7)$ in the range $5 < \ell \leq 6$	each such graph contains a triangle, by Lemma S4.2
$(r, p) = (2, 7)$ in a triangle-free graph	possible with $\ell < 4$; Example S4.3

TABLE 1. Rooted bounds and the candidate pairs above density four. The rows for $4 < \ell \leq 5$ and for $5 < \ell \leq 6$ record necessary pairs only.

The identity $N - 1 = n + q - 2$ gives $(q-1)(N-1) = (q-1)(n-1) + (q-1)^2$, and $(q-3)D_G = (q-3)D + (q-3)$. The terms that do not involve n, D or m are

$$(q-1)^2 - (q-3) - (q-2)(q-3) - 2(q-1) = (q-3)\big((q-1) - (q-1)\big) = 0.$$

What remains is $S(F) \leq (q-1)(n-1) - (q-3)D - m$, which is ROOT($q$). For $q = 4$ this route uses Proposition S1.3. The induction in Theorem S1.2 does not. For $q \geq 5$ the hypothesis is open, so the implication leaves ROOT(q) open.

Conjecture S4.6. *Every connected strong C_4-free graph G that is not a cycle and has $\ell(G) > 5$ has an eligible core H, in the sense of Example S3.5, with*

$$S(G) - S(H) \leq (p - r)\big(n - |H|\big) + |H|.$$

Example S3.5 shows that the budget without a restriction on ℓ is false, and that the Petersen graph is a counterexample with $\ell = 3$. The example does not show that 5 is the correct cutoff, and it does not count eligible cores. A reported count of cores of order at most 12 is not a result of this paper. If the conjecture, the range $\ell \leq 5$, and an induction on the order were all available, REAL-133 would follow. The range $\ell \leq 5$ is not a theorem, so this route is not a proof of REAL-133. Example S3.6 shows that a proof cannot rely only on the scalar constraints listed there.

APPENDIX A. THE FINITE CERTIFICATE

Lemma 5.8 and Lemma 5.10 are proved by finite searches whose completeness arguments are in the article. This section records the audit. The decision array is not printed here, and it is not reconstructed in the text. The companion program `certificates/gen3/verify_gen3.py` is formal submission material. It uses only the Python standard library, requires Python 3.10 or later, and embeds the certificate. From that directory the command is `python3 verify_gen3.py`. The counts below are the output of that program.

The checker labels each new vertex by the current order. It checks that the selected vertex or pair is deficient, visits every child neighbourhood in increasing integer-mask order, and consumes the supplied choices in preorder. A new 4-cycle exists exactly when two members of the new

neighbourhood have a common neighbour in the graph already built. For a new induced P_7, the checker tests six old vertices together with the new vertex and reads their induced edges. At an unrestricted node it compares the labelled graph with $H_{3,2}$ and continues to explore the extensions. A missing choice, or an unused choice, makes the verification fail. The verifier imposes no order cutoff and performs no isomorphism pruning.

The reason a finite tree covers every order is the one given in the proof of Lemma 5.8. If the current induced subgraph H has a vertex v with $L_H(v) < 4$ and v has no neighbour outside H, the degree and the triangle count of v are already final, which contradicts $L \geq 4$ in the finished graph. If two adjacent vertices have degree two in H, at least one has a neighbour outside H. The deficient vertex, or one vertex of that pair, is a set that some new vertex must meet. When H already satisfies both local conditions, the search compares it with $H_{3,2}$. If H is a proper induced subgraph of a connected graph satisfying the hypotheses, some vertex outside H has a neighbour in H, and the node is unrestricted. For a fixed embedding into the target graph, the old neighbourhood of a required new vertex is one of the sets the node enumerates, and the child is kept. An embedded proper prefix therefore extends. Every potential counterexample continues along some branch.

Order	3	4	5	6	7	8	9	10
Visited nodes	1	1	1	1	2	3	5	14
Order	11	12	13	14	15	16	17	
Visited nodes	23	24	23	19	24	20	8	

The full one-triangle tree has 169 labelled nodes and 52 leaves. Four unrestricted nodes occur, all at order 11, and all four represent $H_{3,2}$. The largest order reached is 17. That value is an output of the search.

The second certificate is the search in the proof of Lemma 5.10. It uses different initial graphs and different decisions. At each node the verifier checks a single vertex of current deletion loss less than 4 and explores every legal extension that meets that vertex. The four rows are the shared-vertex configuration, the single joining edge, and the joining paths of lengths two and three.

Initial configuration	Initial order	Visited nodes	Largest order
Shared vertex	5	36	17
One joining edge	6	31	15
Joining path of length two	7	46	11
Joining path of length three	8	5	10

These searches visit 118 nodes in all, and every branch closes. An induced embedding into a graph with $L \geq 4$ at every vertex would continue at every node, which is the local conclusion of Lemma 5.10.

Appendix B. A conditional reduction for GEN_3

Theorem 5.9 is the bound $S \leq 3n$ for every connected strong C_4-free P_7-free graph, with equality only for the Petersen graph. Its proof is the triangle-free case, Theorem 5.4, together with Lemma 5.8. The argument below is not used in that proof. It records the shape a minimum counterexample would have had, and the three local suppositions which would also have implied the bound. Those suppositions are not needed. Proposition 5.16 is the intermediate edge-deficit estimate $3 < \ell \leq 4 - 16/n$ on the cell $r = 3$, $p = 6$, $T \geq 2$ and $n \geq 17$. Theorem 5.9 shows that the cell is empty, and Corollary 5.19 is REAL-133 for every connected graph with $\ell \leq 4$.

Theorem SB.1. *Let G be a counterexample to* GEN_3 *of minimum order: G is connected and strong C_4-free, $p(G) \le 6$, $S(G) \ge 3n+1$, and n is minimum. Then G is 2-connected; $S(G) - S(G-U) \ge 3|U|+1$ for every nonempty proper $U \subset V(G)$; $2\lambda(v) - t(v) \ge 4$ for every v, so $\lambda(v) \ge 2$; and every vertex with $\lambda(v) = 2$ has degree 2 and nonadjacent neighbours, and no two vertices of degree 2 are adjacent.*

Proof. Since $S \le 2 < 3n$ for $n \le 2$, one has $n \ge 3$. Let v be a cut vertex. For each component C of $G-v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ with end v. Two such paths from different components form an induced path on $\rho_C + \rho_{C'} - 1 \le 6$ vertices, so some component C_1 has $\rho_{C_1} \le 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Both graphs are connected, smaller than G, and P_7-free. Lemma 5.3 gives $S(G_1) \le 2(n_1 - 1)$, and minimality gives $S(G_2) \le 3n_2$. By Lemma 2.3, $S(G) \le 2(n_1-1) + 3n_2 = 3n - (n_1 - 1) < 3n$.

Every component H of $G - U$, for U nonempty and proper, is connected, smaller, and P_7-free, so $S(H) \le 3|H|$. Summing, $S(G-U) \le 3(n - |U|)$, and $S(G) \ge 3n+1$ gives the deletion bound. The case $U = \{v\}$ and Lemma 2.2 give $2\lambda(v) - t(v) \ge 4$. If $\lambda(v) = 2$, then $t(v) = 0$, so $d(v) = 2$ and the neighbours are nonadjacent. Every degree-2 vertex has $\lambda = 2$ and hence lies in no triangle. If $u \sim w$ both have degree 2, with other neighbours x and y, then $x \ne y$ because $t(u) = 0$. The set $\{u\}$ is a unit of x and $\{w\}$ is a unit of y, so Lemma 2.2 with $U = \{u, w\}$ gives loss $2 + 2 + 1 + 1 = 6 < 7$. □

The same mass bound for a disconnected graph is the sum of the bounds on its components. An isolated vertex has $S = 0$. Call a graph *admissible* if it is strong C_4-free and P_7-free, $L(v) \ge 4$ for every vertex, and no two adjacent vertices have degree 2. A minimum counterexample is admissible. In an admissible graph, $\lambda(v) = 2$ forces $t(v) = 0$ and $d(v) = 2$. Write B for the vertices of degree at least 3, the *branch vertices*, and Z for the degree-2 vertices, the *ears*. Put $D = G[B]$, $d_D(v) = |N(v) \cap B|$ and $q(v) = |N(v) \cap Z|$. All triangles lie in D.

Lemma SB.2. *In an admissible graph, $q(v) \le 4$ for every vertex v.*

Proof. Let u_i be ears at v with other neighbours a_i. The a_i are distinct, nonadjacent to v, and of degree at least 3. Put $E_i = N(a_i) \setminus \{u_i\}$. Then $|E_i| \ge 2$ and $E_i \cap N[v] = \emptyset$.

If $a_i \not\sim a_j$, then $E_i = \{w, p\}$ and $E_j = \{w, q\}$ with $pq \in E$, $w \not\sim p$ and $w \not\sim q$. Indeed two common neighbours of a_i and a_j are forbidden, so $|E_i \cap E_j| \le 1$. Since $|E_i| \ge 2$, the private part $E_i \setminus E_j$ is nonempty, and likewise $E_j \setminus E_i$. If p lies in the first private part and q in the second with $p \not\sim q$, the path $pa_iu_ivu_ja_jq$ is induced. Thus the private parts are complete to each other. If one of them contained two vertices, those two vertices and one vertex of the other private part would form a 4-cycle with the corresponding a-vertex. Both private parts therefore have size one, and $|E_i \cap E_j| = 1$. An edge from the common vertex to a private vertex is another 4-cycle. In particular a_i and a_j have degree 3 and lie in no triangle.

No three of the vertices a_i are pairwise nonadjacent. Otherwise the sets E_i meet pairwise in a single vertex. If the three intersections are the three pairs of a triple, the claim forces a triangle on that triple, and one a_i together with two vertices of the triple forms a 4-cycle. Otherwise $E_i = \{w, p_i\}$ with $p_1p_2p_3$ a triangle and w nonadjacent to each p_i, and $vu_1a_1wa_2p_2p_3$ is an induced P_7.

Five ears are impossible. The graph F they determine on $\{a_1, \dots, a_5\}$ has independence number at most two. If some vertex a of F is adjacent to the other four, those four induce a matching. The independence number forces that matching to be perfect, say $\{b_1b_2, c_1c_2\}$. Then $b_1 \not\sim c_1$, so the preceding paragraph says that b_1 lies in no triangle, but ab_1b_2 is a triangle. Otherwise every vertex of F has a non-neighbour, hence has degree 3 in G and lies in no triangle. One neighbour of each such vertex is its ear, so F has maximum degree at most two. It is triangle-free and has no 4-cycle, and its independence number is at most two, so F is a 5-cycle. Label that cycle $a_1a_2a_3a_4a_5a_1$, and write u_i for the ear joining v to a_i. The seven vertices $u_5, v, u_1, a_1, a_2, a_3, a_4$,

in that order, induce a P_7: each u_i has neighbours v and a_i only, v is nonadjacent to every a_j, and the only edges of the cycle among a_1, a_2, a_3, a_4 are the three path edges. □

Proposition SB.3. *Let G be admissible, with B, Z, D, d_D and q as above. Suppose that $d_D(v) \le 3$ for every v, that $d_D(v) = 3$ implies $q(v) = 0$, and that $d_D(v) = 2$ with the two neighbours in B nonadjacent implies $q(v) \le 2$. Then $S(G) \le 3n$.*

Proof. Ears have nonadjacent neighbours and Z is an independent set, so $S(G) = S(D) + 4|Z|$. Since $|Z| = \frac{1}{2}\sum_{v\in B} q(v)$ and $n = |B| + |Z|$,

$$S - 3n = \sum_{v\in B}\Big(d_D(v) - t_D(v) - 3 + \tfrac{1}{2}q(v)\Big).$$

Lemma SB.2 gives $q(v) \le 4$. If $d_D(v) = 2$ and $t_D(v) = 0$, the two neighbours in B are nonadjacent, for an edge between them would be a triangle. The three suppositions and the ear bound make every summand nonpositive: the summand is at most -1 when $(d_D, t_D) = (0, 0)$, at most 0 when $(d_D, t_D) = (1, 0)$, at most 0 when $(d_D, t_D) = (2, 0)$, at most 0 when $(d_D, t_D) = (2, 1)$, and equal to $-t_D(v) \le 0$ when $d_D(v) = 3$. The suppositions exclude $d_D(v) \ge 4$. □

The three suppositions in Proposition SB.3 are not proved in this paper, and they are not used for Theorem 5.9. Theorem SB.1 shows that a minimum counterexample to GEN_3, if one existed, would be admissible, so those suppositions would also have implied the bound. The proof in the article does not pass through them.

Acknowledgments

The authors are solely responsible for the content of this article. Language-model tools were used in the preparation of drafts of the manuscript.

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